

NIST Cybersecurity Framework to respond to a security incident

Scenario

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DDoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a distributed denial of service (DDoS) attack.

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below.

Role: Cybersecurity Analyst

Company: (Multimedia company offering web design, graphic design, and social media marketing)

Incident Type: Distributed Denial of Service (DDoS) Attack

Summary	The company experienced a distributed denial of service (DDoS) attack that caused a major disruption in network services for two hours. The attack was caused by a flood of ICMP packets sent through an unconfigured firewall, which allowed the attacker to overwhelm internal network traffic. The incident response team blocked incoming ICMP packets, took non-critical services offline, and restored critical services to resume normal operations.
Identify	A DDoS attack using ICMP flooding targeted the company's internal network. The attacker took advantage of a firewall that wasn't properly configured. This allowed them to send large amounts of ICMP packets, making internal network resources unreachable. All systems connected to the network were impacted

	and had to be restored.
Protect	The cybersecurity team added a firewall rule to limit incoming ICMP packet rates. They also set up an IDS/IPS system to block suspicious ICMP traffic. These actions will help prevent future ICMP flood attacks and secure the network better.
Detect	To catch future attacks early, the team enabled source IP address verification on the firewall to identify spoofed IPs. They also installed network monitoring software to spot unusual traffic patterns and detect potential DDoS behavior faster.
Respond	In future incidents, the team will isolate affected systems right away to reduce damage. They will prioritize restoring critical services, review network logs for suspicious activity, and report the incident to management. If needed, they will inform legal authorities too.
Recover	restoring access to all network services in stages. First, non-critical services are stopped to lower internal traffic. Then, critical services are restored. Once the attack slows down, non-critical services can safely be brought back online. The firewall will continue blocking external ICMP floods to help prevent another outage.

Reflections/Notes: This incident showed how important it is to keep firewalls properly configured and to monitor traffic for unusual activity. After applying the NIST Cybersecurity Framework, the company now has better protection, faster detection tools, and a clear response and recovery plan for future events.